

ZANESVILLE, OH, USA

WMO: 724286

Lat: 39.944N

Lon: 81.892W

Elev: 268

StdP: 98.14

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93824

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.9	-12.7	-20.5	0.6	-14.7	-17.6	0.8	-11.7	10.4	-0.1	9.1	0.4	2.3	260	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	32.1	23.2	30.8	22.7	29.5	21.9	24.7	29.9	23.9	28.6	23.1	27.5	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	18.4	27.4	22.4	17.6	26.5	21.7	16.9	25.7	76.1	30.2	72.6	28.7	69.7	27.7	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.6	6.8	DB	-20.5	33.8	4.4	1.7	-23.7	35.1	-26.3	36.1	-28.8	37.0	-32.0	38.3	(4)
(5)			WB	-20.9	26.1	4.4	0.9	-24.0	26.7	-26.6	27.2	-29.0	27.7	-32.2	28.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.5	-1.5	0.3	5.0	11.5	16.7	21.2	23.0	22.4	18.7	12.3	6.1	1.3	(6)
(7)		DBStd	9.96	7.00	6.37	6.05	5.13	4.41	3.32	2.73	2.70	3.87	4.72	5.21	5.64	(7)
(8)		HDD10.0	1312	363	278	179	43	3	0	0	0	1	29	139	276	(8)
(9)		HDD18.3	2981	616	507	414	211	86	13	1	3	41	196	366	528	(9)
(10)		CDD10.0	1848	6	6	24	87	211	336	404	383	262	100	23	6	(10)
(11)		CDD18.3	476	0	0	1	6	35	99	147	128	53	8	0	0	(11)
(12)		CDH23.3	4113	0	1	11	74	343	840	1257	1044	469	72	2	0	(12)
(13)		CDH26.7	1243	0	0	1	9	79	257	422	326	138	10	0	0	(13)
(14)	Wind	WSAvg	2.8	3.5	3.4	3.4	3.4	2.7	2.4	2.1	1.9	2.0	2.5	3.0	3.2	(14)
(15)	Precipitation	PrecAvg	985	60	56	86	91	101	99	108	95	77	60	79	70	(15)
(16)		PrecMax	1383	151	106	251	174	202	330	207	270	158	141	339	185	(16)
(17)		PrecMin	674	12	8	25	20	39	17	31	16	9	5	18	25	(17)
(18)		PrecStd	155	31	28	45	41	44	63	54	57	40	33	52	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.9	19.9	25.0	28.0	30.6	32.7	33.9	33.1	32.5	28.3	22.6	18.4	(19)
MCWB			13.8	13.4	16.2	17.4	21.5	22.9	24.6	24.1	21.9	20.2	15.6	14.5	(20)	
2%		DB	15.0	16.1	21.4	25.6	28.8	31.1	32.1	31.3	30.3	25.5	19.6	15.1	(21)	
		MCWB	12.5	11.3	13.9	16.2	20.1	22.6	23.8	23.1	21.5	18.2	14.1	12.1	(22)	
5%		DB	12.1	13.0	18.2	23.3	27.3	29.6	30.5	29.9	28.0	23.1	17.3	12.8	(23)	
		MCWB	10.0	9.5	12.1	15.0	19.5	21.8	23.1	22.4	20.3	17.0	12.5	10.2	(24)	
10%		DB	8.6	9.9	15.2	20.9	25.2	28.0	29.1	28.5	26.1	20.8	14.9	10.1	(25)	
		MCWB	6.1	6.7	10.9	13.6	18.5	21.1	22.3	21.7	19.5	16.1	11.2	7.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	15.0	16.8	19.5	22.9	24.8	25.8	25.7	23.9	21.7	17.4	15.3	(27)
MCD B			16.9	19.0	23.3	25.9	28.6	30.4	31.7	30.9	29.4	27.4	20.7	17.7	(28)	
2%		WB	12.8	12.5	15.0	17.5	21.5	23.6	24.9	24.6	22.7	19.9	15.2	12.8	(29)	
		MCD B	14.8	15.0	19.7	23.0	26.7	28.8	30.4	29.4	27.5	23.5	18.5	14.6	(30)	
5%		WB	10.3	9.9	13.3	16.2	20.4	22.8	24.0	23.7	21.7	18.2	13.5	10.6	(31)	
		MCD B	12.0	12.7	17.2	21.2	25.4	28.0	28.7	27.9	26.0	21.8	16.4	12.4	(32)	
10%		WB	6.4	6.8	11.2	14.9	19.4	22.1	23.2	22.7	20.7	16.5	11.6	7.8	(33)	
		MCD B	8.3	9.7	14.8	19.4	23.9	26.8	27.5	26.5	24.4	20.0	14.7	10.1	(34)	
(35)	Mean Daily Temperature Range	MDBR		8.5	9.3	10.7	12.3	11.8	11.1	10.9	11.2	11.9	11.6	10.1	8.0	(35)
5% DB		MCD BR		10.4	13.6	15.2	15.5	14.2	12.7	12.1	12.7	13.9	14.4	13.5	11.3	(36)
		MCW BR		8.5	10.1	9.3	8.5	7.0	5.8	5.2	5.5	6.3	8.0	9.1	9.3	(37)
5% WB		MCD BR		9.8	12.4	12.7	12.6	11.8	10.9	10.6	10.6	11.1	11.9	11.1	10.4	(38)
		MCW BR		8.5	10.3	8.6	7.8	6.3	5.5	5.0	5.2	5.5	7.4	8.7	9.5	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.308	0.326	0.351	0.390	0.432	0.459	0.469	0.461	0.399	0.364	0.323	0.304	(40)
taud		2.452	2.387	2.380	2.290	2.191	2.159	2.148	2.153	2.304	2.402	2.494	2.499	(41)		
Ebn at Noon		858	890	903	884	850	825	812	807	839	834	835	837	(42)		
Edh at Noon		79	97	109	128	144	149	150	145	117	95	76	71	(43)		
(44)	All-Sky Solar Radiation	RadAvg		1.61	2.41	3.45	4.65	5.42	5.93	5.78	5.37	4.39	3.00	2.04	1.38	(44)
(45)		RadStd		0.15	0.34	0.32	0.44	0.45	0.43	0.36	0.30	0.41	0.29	0.28	0.16	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+107	N/A		
(47)	Regional (42 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+129	+69		

Nomenclature: See separate page